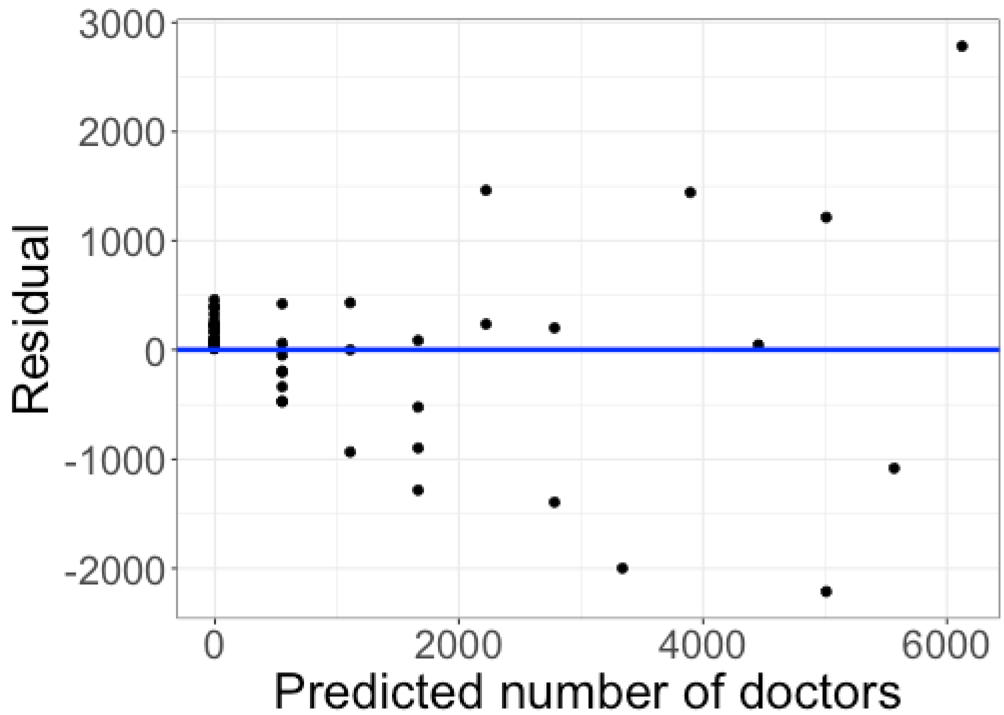
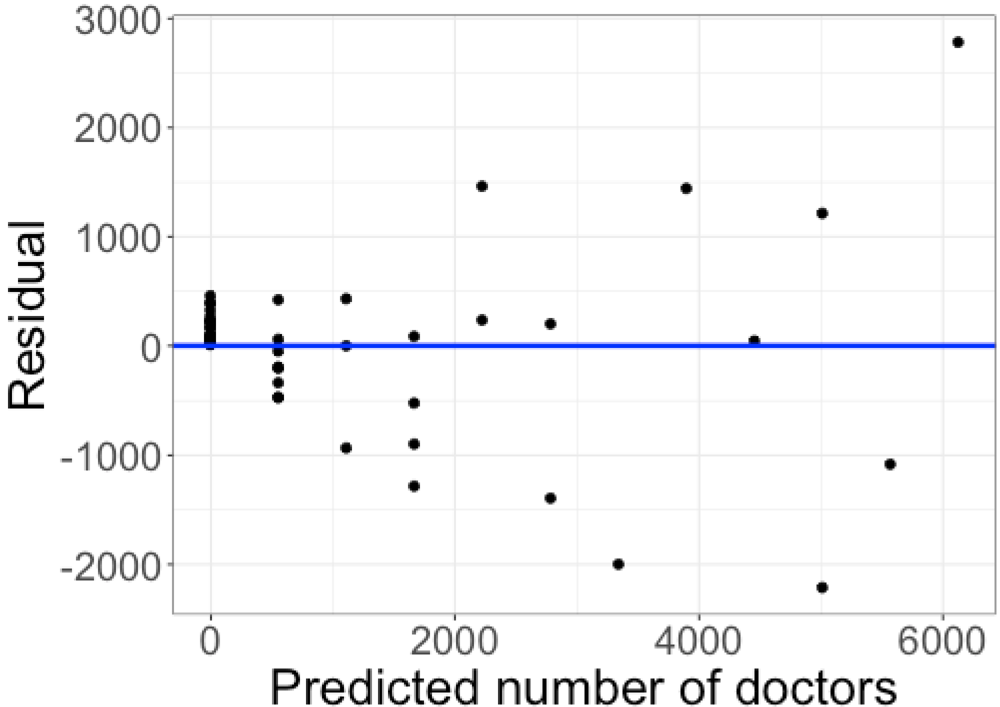
Nonlinear relationships and transformations

Last time



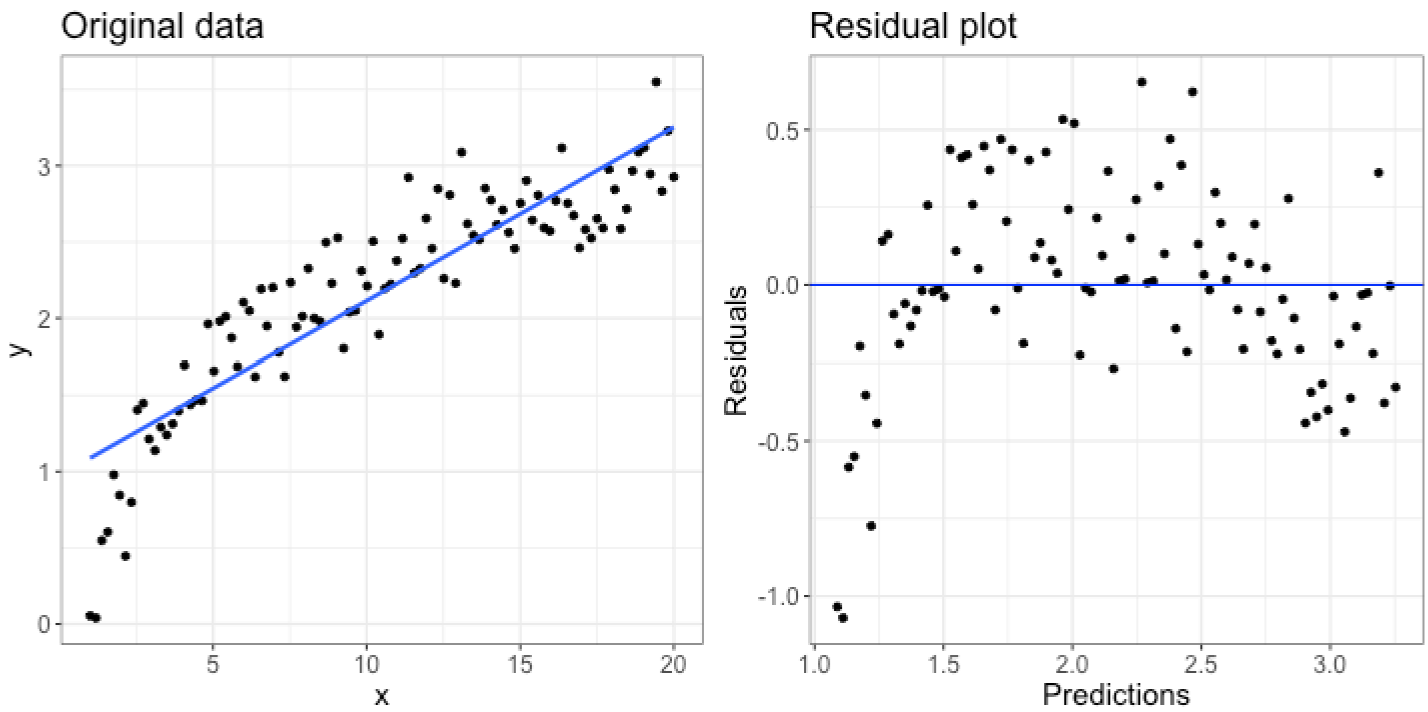
> Do the shape and constant variance assumptions look reasonable?

Today



Today: handling non-linear relationships

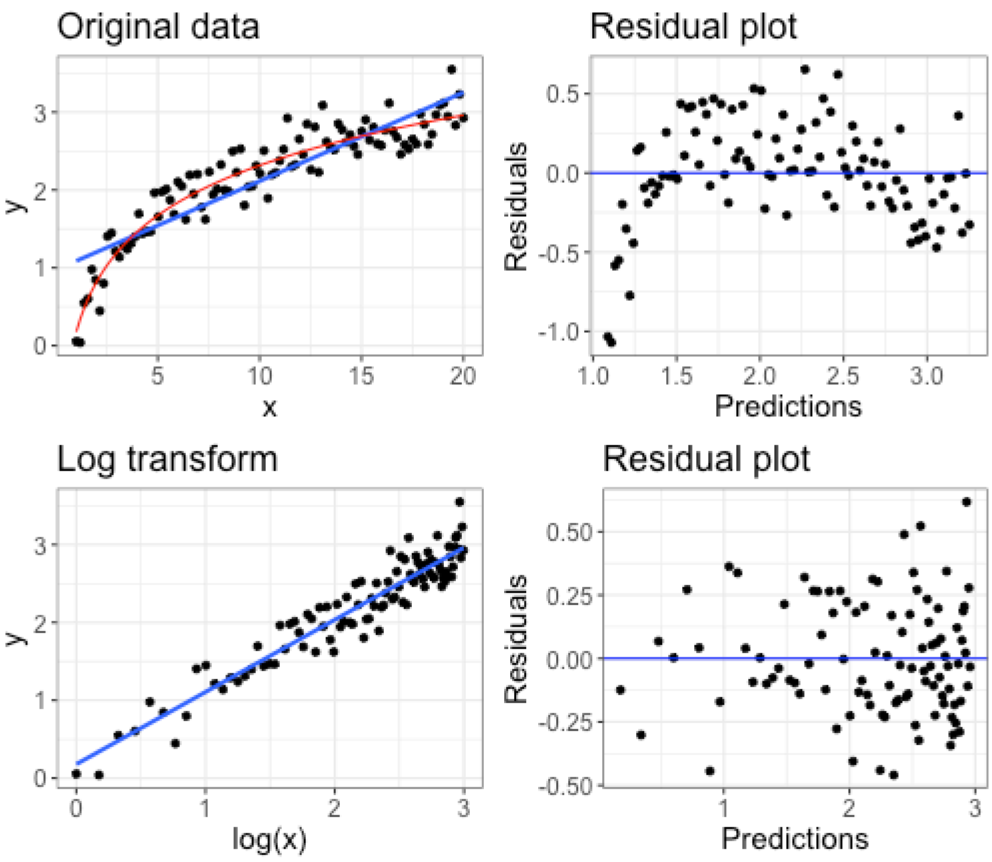
Example



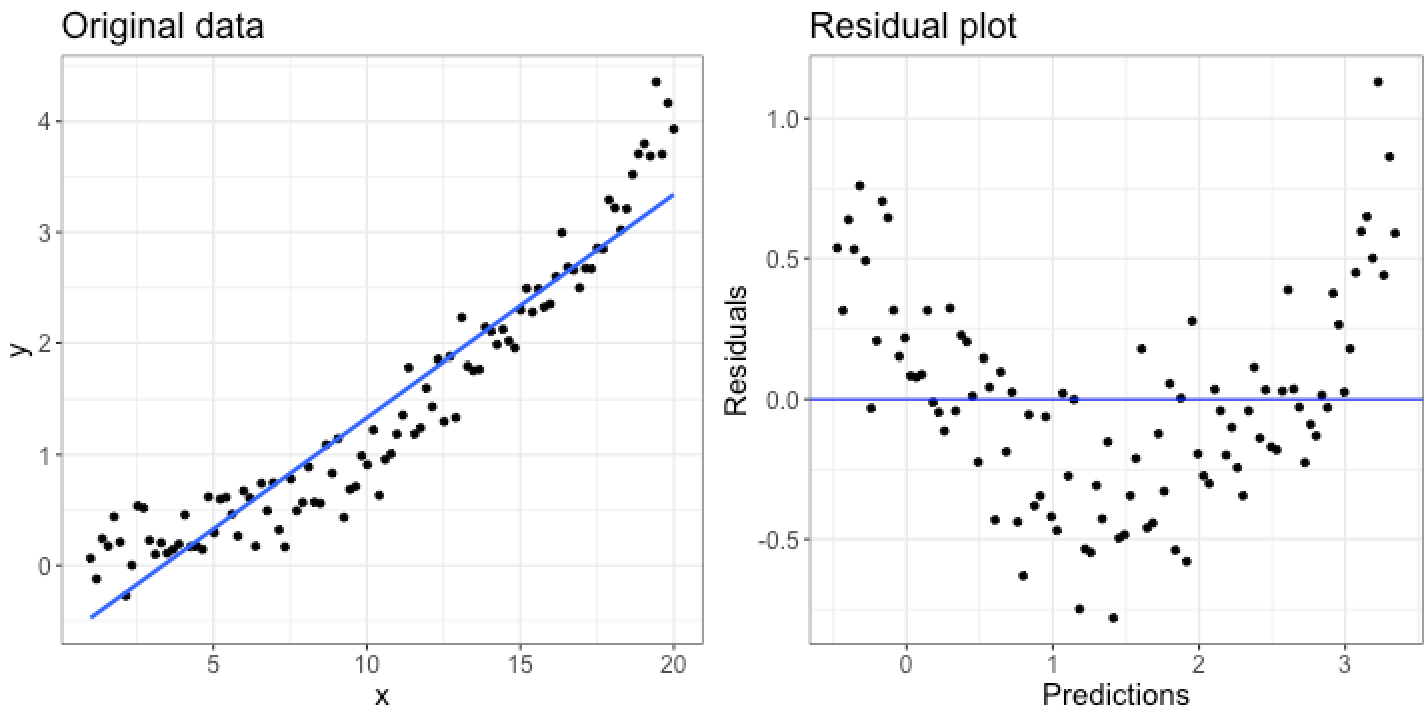
> The relationship between x and y does not appear linear. How could we model that relationship instead?

Example: log transform

$$Y = \beta_0 + \beta_1 \log(X) + \varepsilon$$



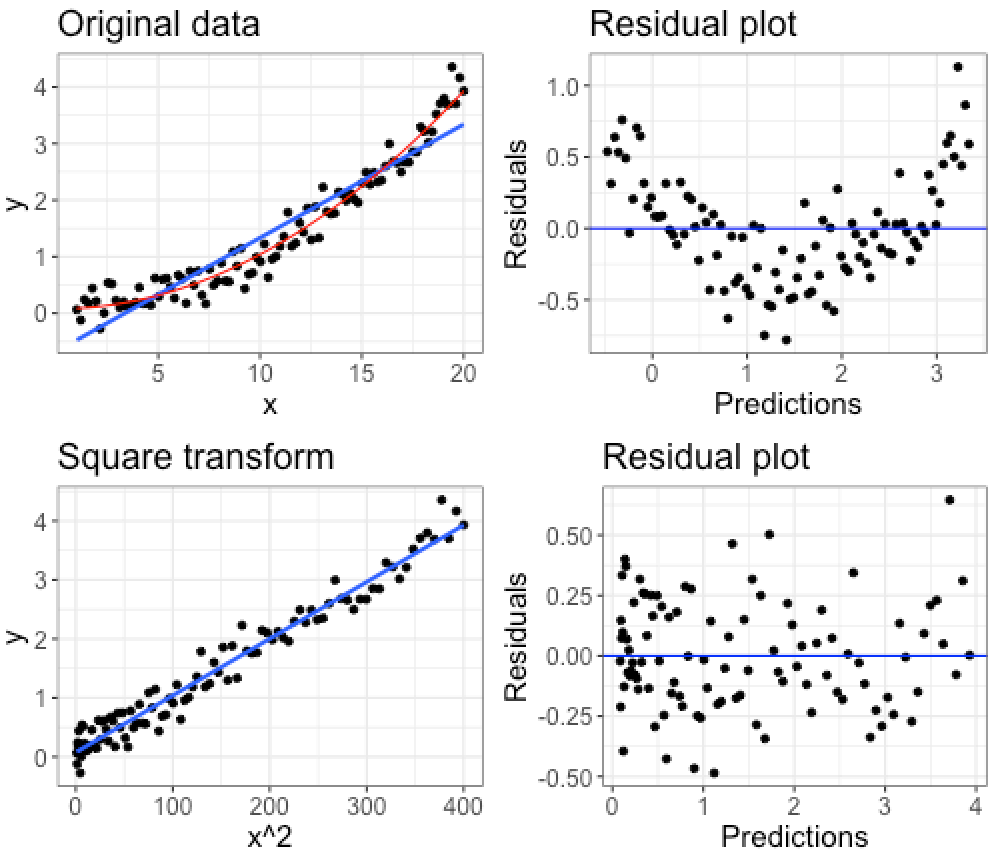
Example



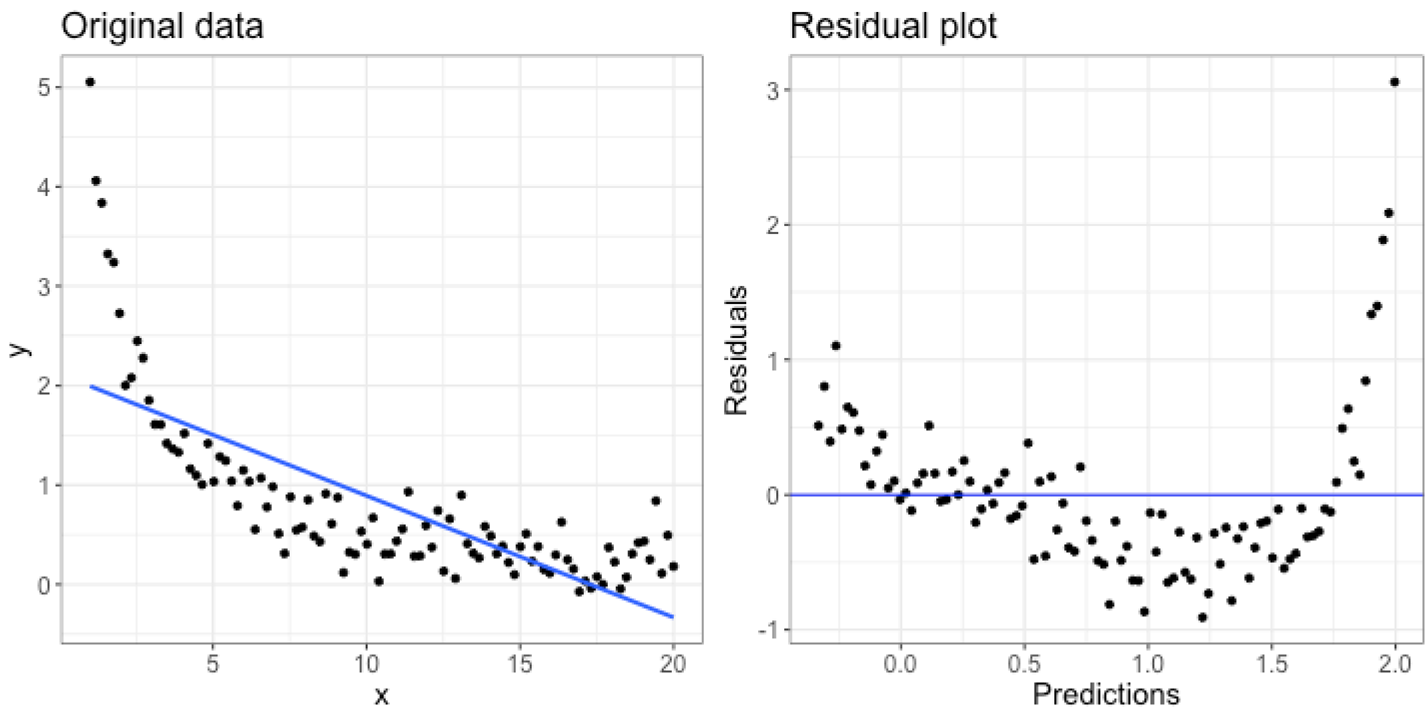
> The relationship between x and y does not appear linear. How could we model that relationship instead?

Example: squaring X

$$Y = \beta_0 + \beta_1 X^2 + \varepsilon$$



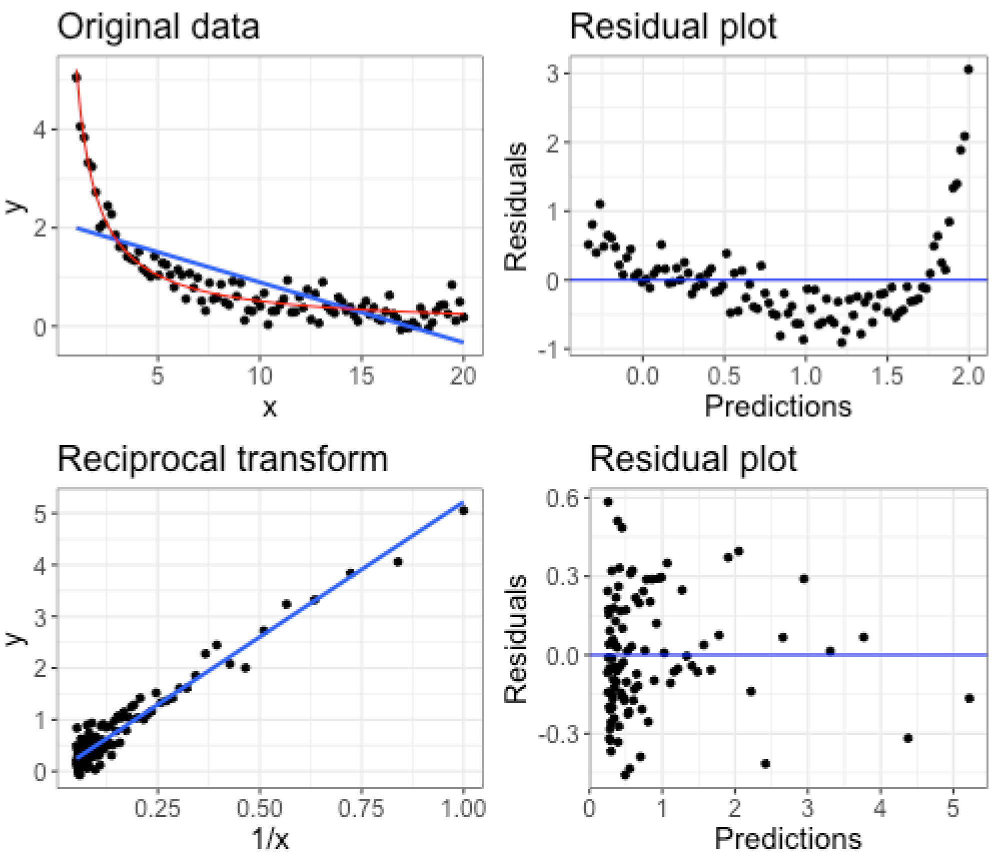
Example



> The relationship between x and y does not appear linear. How could we model that relationship instead?

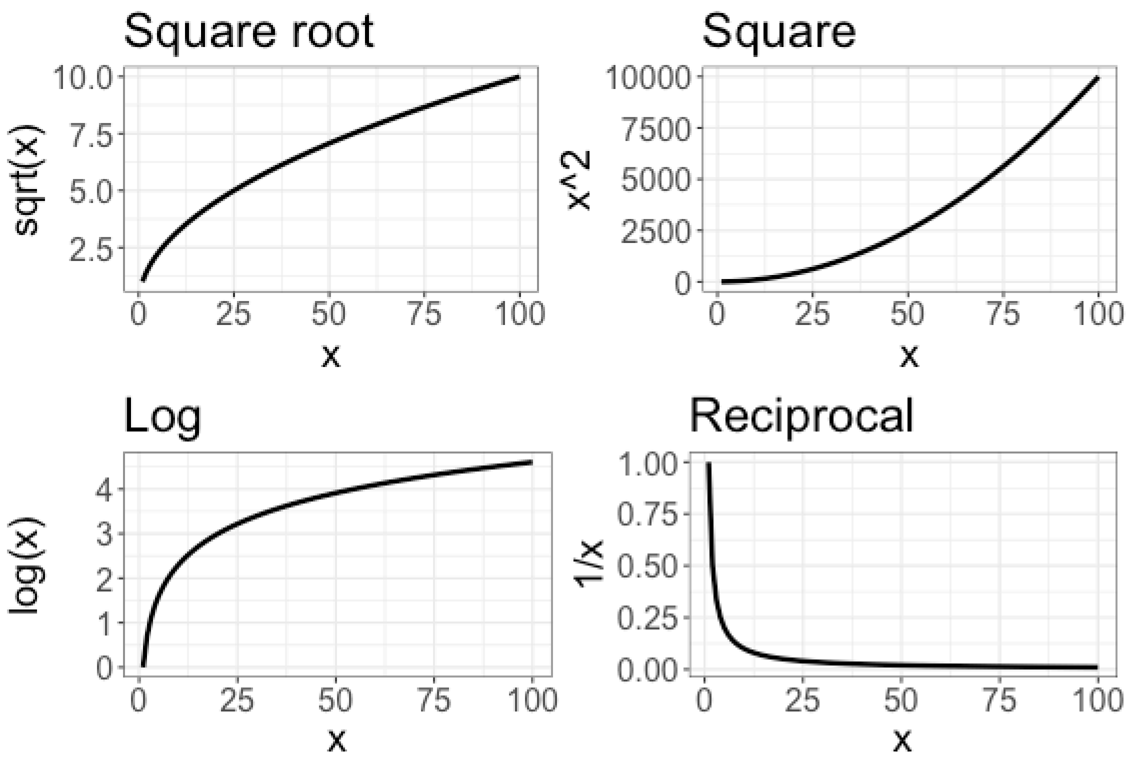
Example: reciprocal transform

$$Y = \beta_0 + \beta_1\,(1/X) + \varepsilon$$

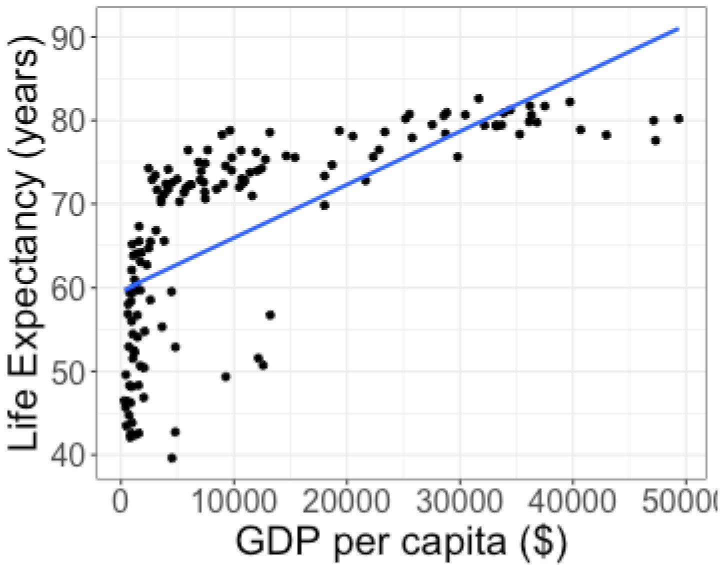


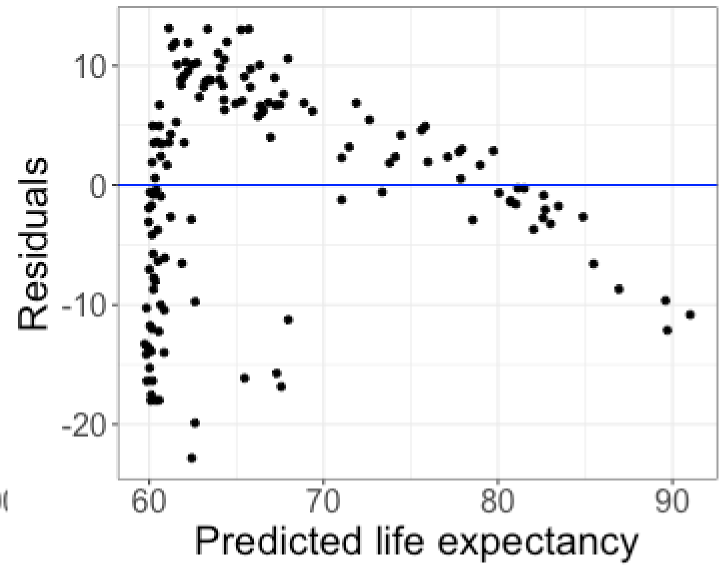
Choosing transformations

To model nonlinear relationships, try transformations for x that looks like the relationship



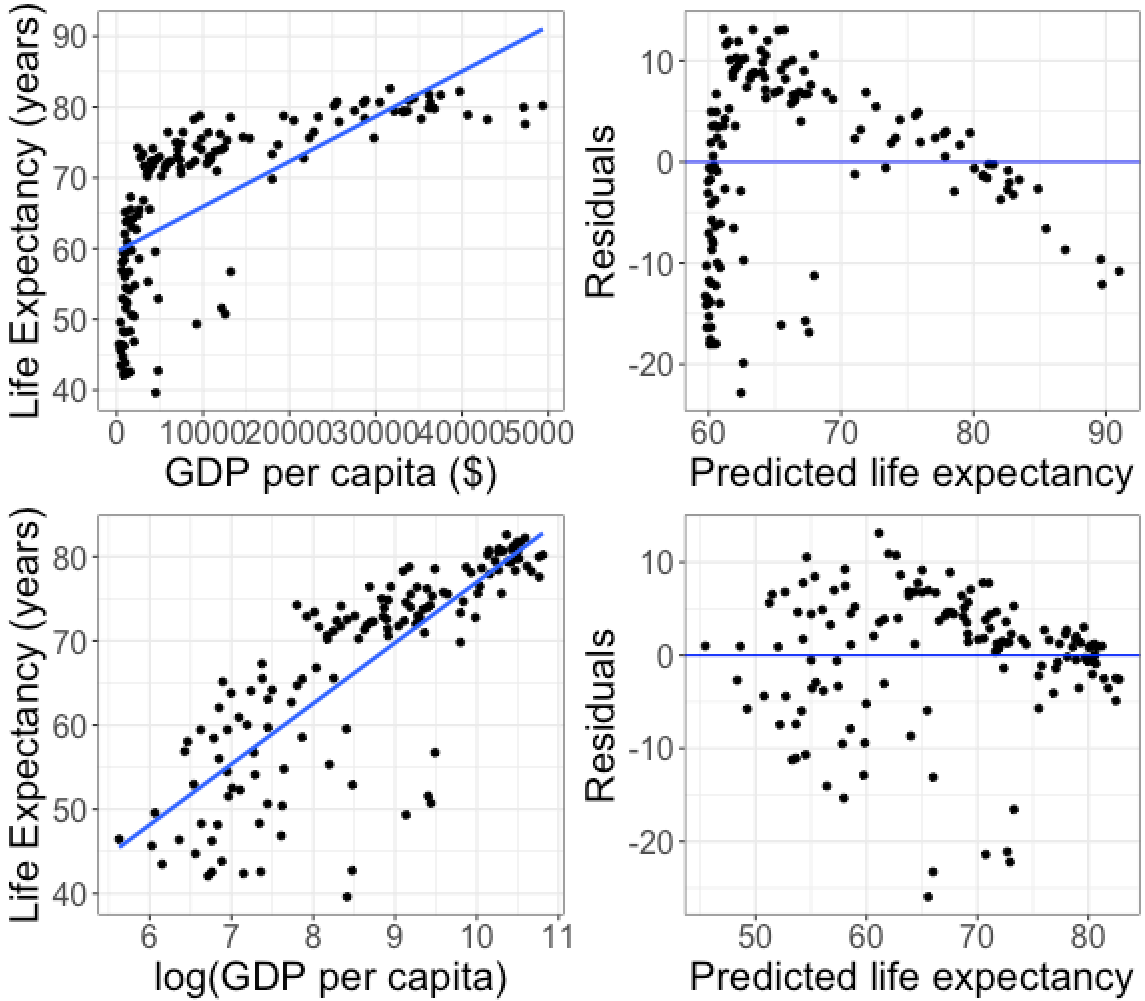
Example



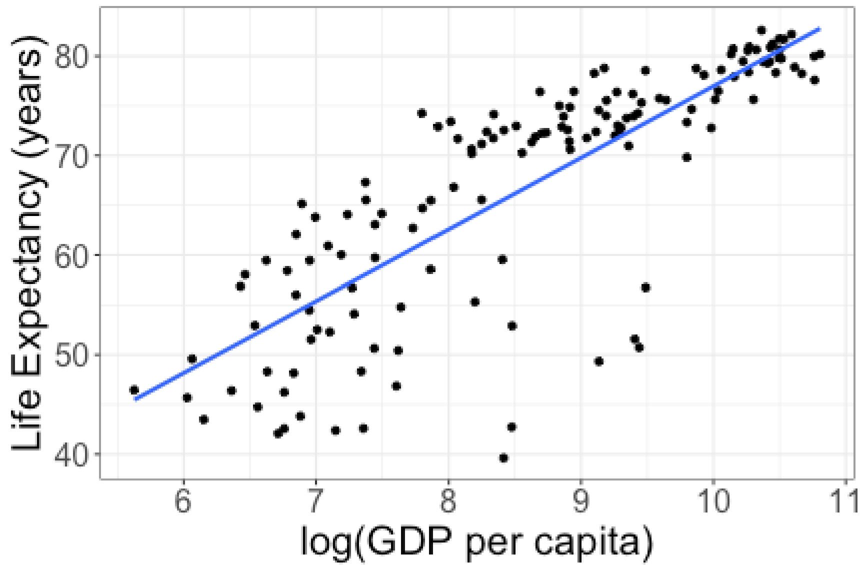


> What transformation could I try for this data?

Log transformation



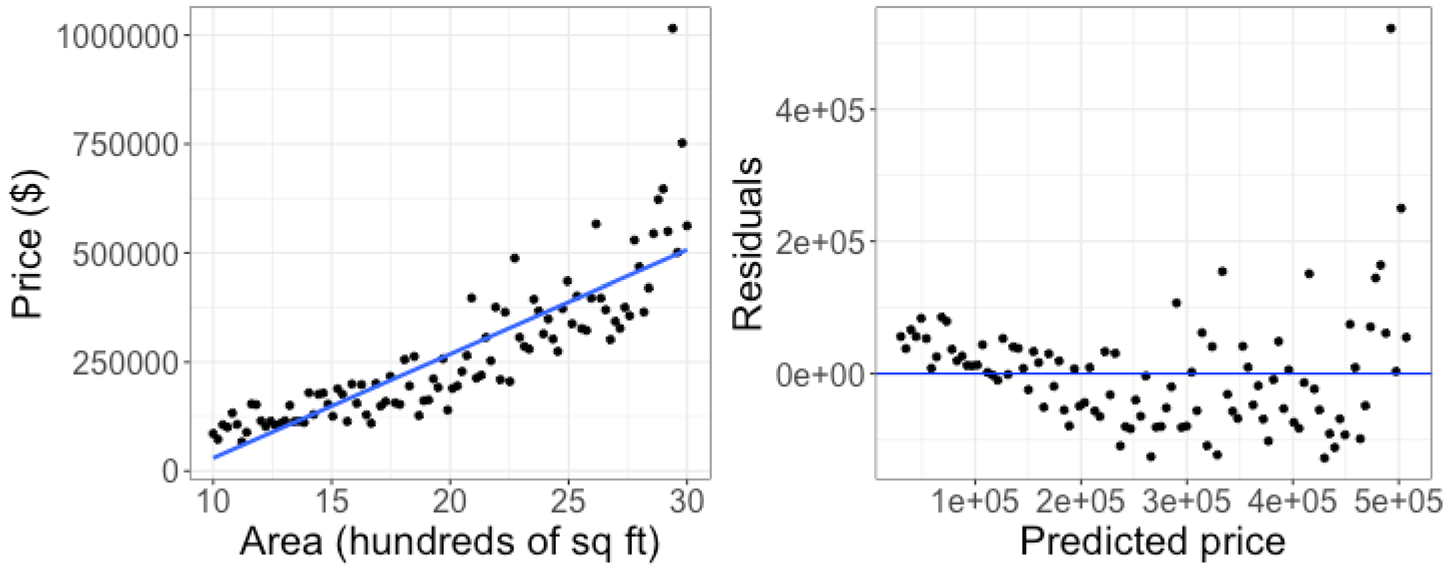
Log transformation



$$\widehat{\text{LifeExpectancy}} = 4.95 + 7.20 \log(\text{GDP per capita})$$

What life expectancy would I predict for a GDP per capita of \$20000?

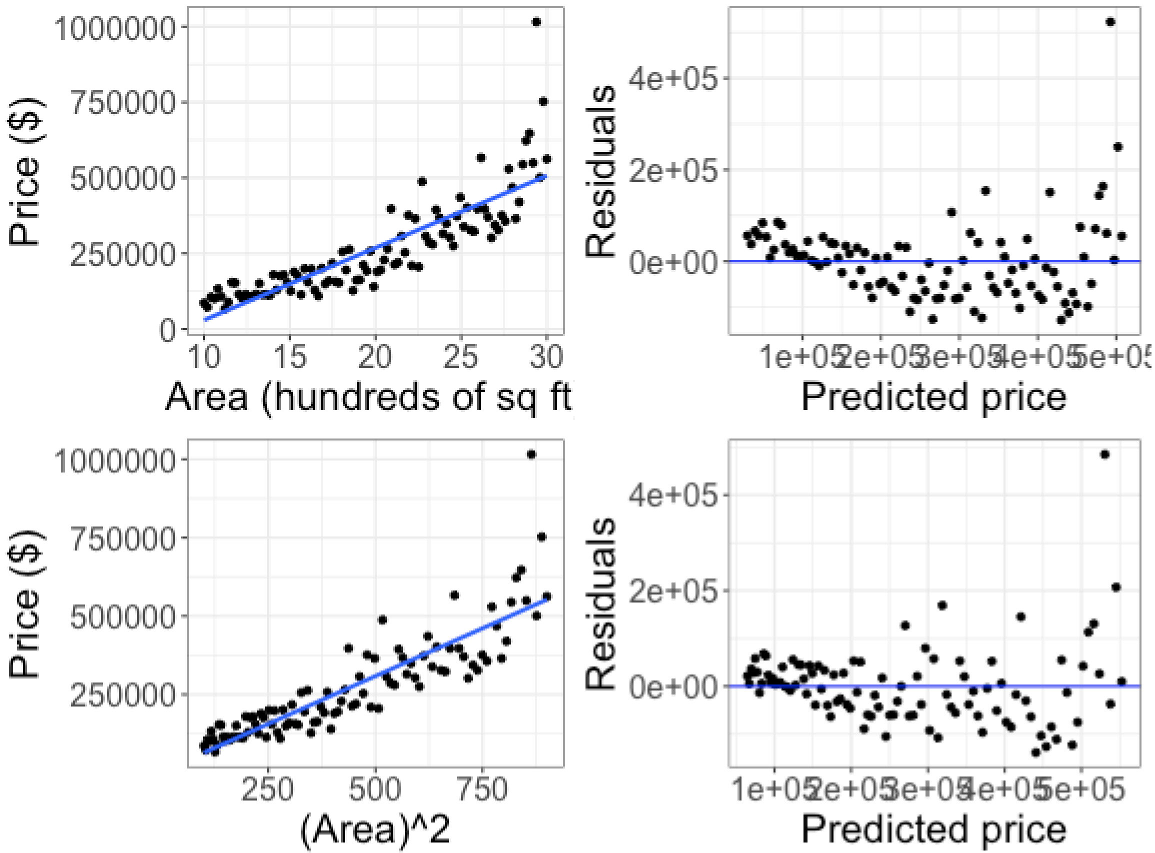
Example



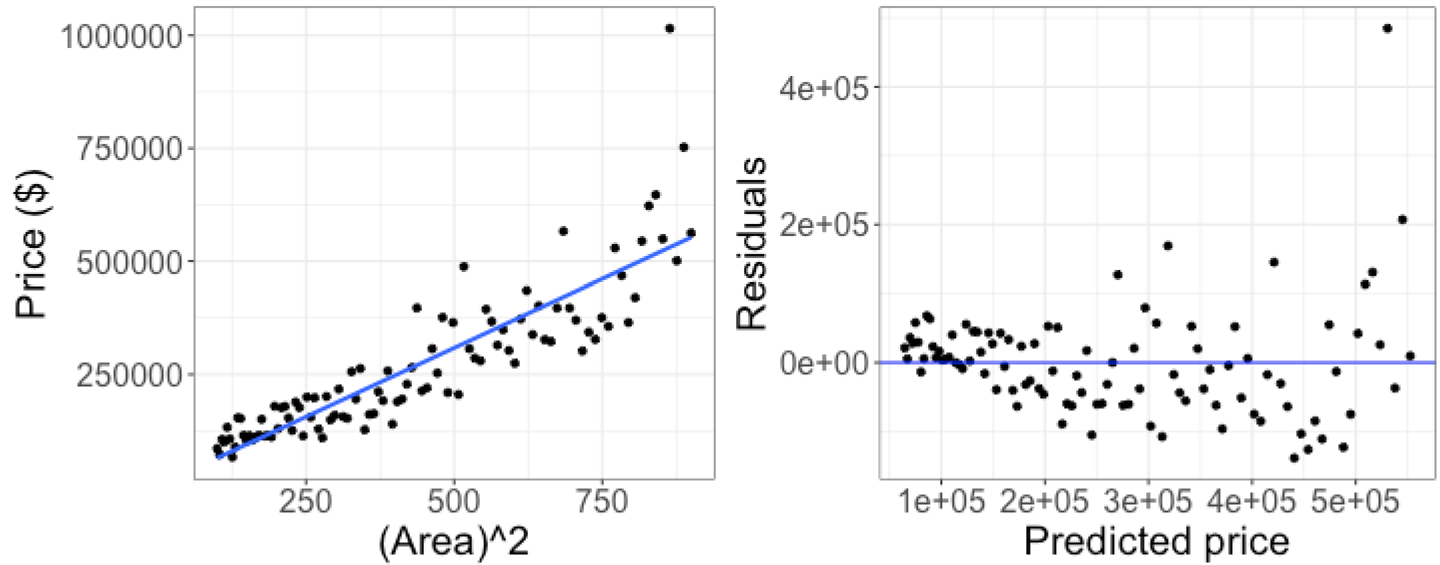
What transformation could I try for this data?

Trying squaring X

Model: $Price = \beta_0 + \beta_1(Area)^2 + \varepsilon$



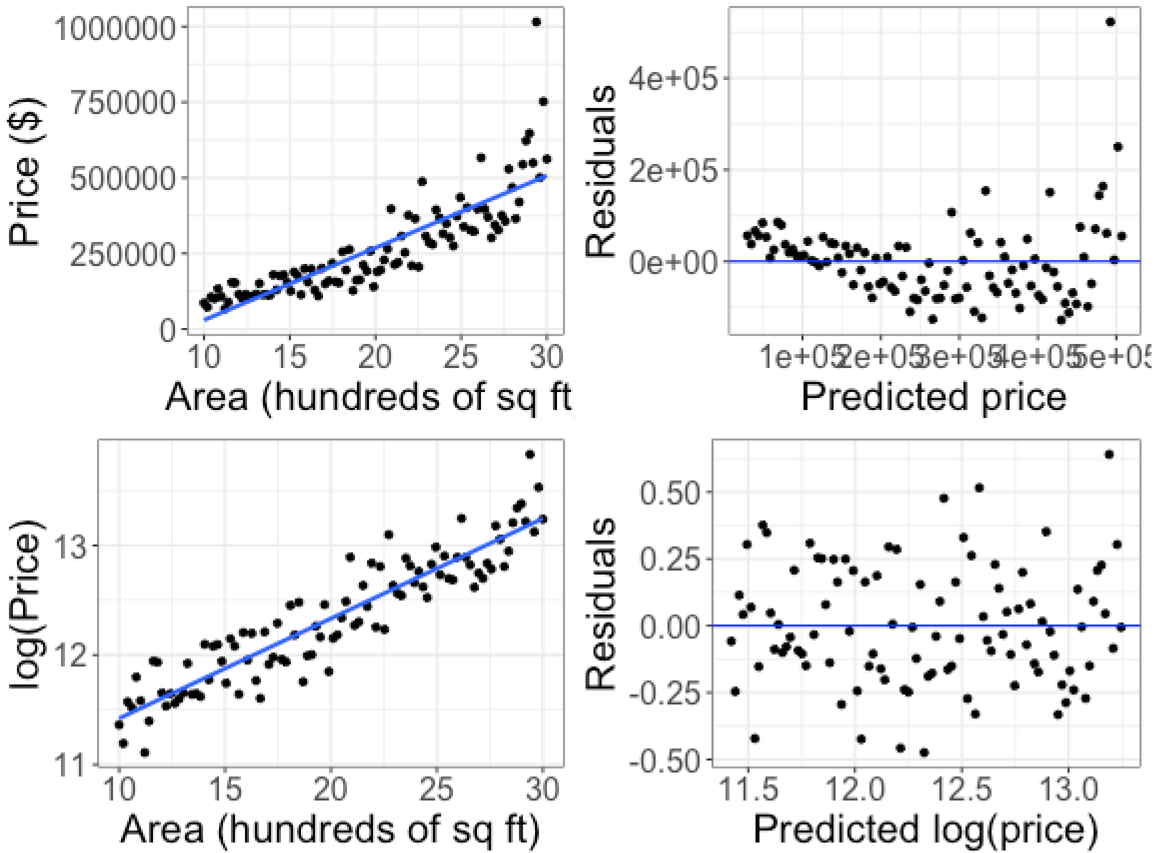
Trying squaring X



What do you notice about the residual plot after trying the transformation?

Another attempt: transforming the response!

Model: $\log(Price) = \beta_0 + \beta_1 Area + \varepsilon$



Log transformed response

$$\widehat{\log(\text{Price})} = 10.51 + 0.091 \text{ Area}$$

- Here Price is in \$, Area is in hundreds of square feet

> How would I calculate the predicted *price* (in \$) for a house of 2000 square feet?

Class activity

Work on the handout, then we will discuss as a class

Class activity

$$\widehat{\text{LifeExpectancy}} = 4.95 + 7.20 \log(\text{GDP per capita})$$

What is the predicted price for GDP per capita of $20200?

Class activity

$$\widehat{\text{LifeExpectancy}} = 4.95 + 7.20 \log(\text{GDP per capita})$$

- When GDP per capita = 20000: $\widehat{\text{LifeExpectancy}} = 76.255$
- When GDP per capita = 20200: $\widehat{\text{LifeExpectancy}} = 76.327$

Change:

Class activity

$$\widehat{\log(\text{Price})} = 10.51 + 0.091\ \text{Area}$$

- Price in \$, Area in hundres of sq ft

> What is the predicted price for a house of 2100 square feet?

Class activity

$$\widehat{\log(\text{Price})} = 10.51 + 0.091 \text{ Area}$$

- Price in $, Area in hundres of sq ft
- When Area = 20: $\widehat{\text{Price}} = 226386$
- When Area = 21: $\widehat{\text{Price}} = 247954$

Change:

Class activity

$$\widehat{\log(\text{Price})} = 10.51 + 0.091 \text{ Area}$$

- Price in \$, Area in hundres of sq ft
- When Area = 20: $\widehat{\text{Price}} = 226386$
- When Area = 21: $\widehat{\text{Price}} = 247954$

Change: $\dfrac{247954}{226386} = 1.095 = e^{0.091}$